
Major Project on Predicting rate of Employee Attrition

INTERNSHIP PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of the degree

Of

**BACHELOR of TECHNOLOGY
DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCES**

By

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07501172021**

Guided by

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Research Associate
Centre of Excellence -Artificial Intelligence, IGDTUW**

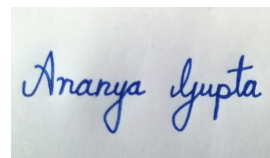


**INDIRA GANDHI DELHI TECHNICAL UNIVERSITY
FOR WOMEN
NEW DELHI - 110006**

- 5th June,2022 to 16th July, 2022 -

CERTIFICATE

I, Ananya Gupta, certify that the Industrial Project Report entitled “Major Project on Predicting rate of Employee Attrition” is done by me and it is authentic work carried out by me at the Department of Artificial Intelligence and Data Sciences, IGDTUW. For this project, no work has been submitted before for any degree or diploma of the award, to the best of my knowledge and belief.



(Ananya Gupta)

Signature of the student

Date: 07/11/2022

Certified that the project report entitled “Major Project on Predicting rate of Employee Attrition & Minor project on Honey Production Analysis” done by the above student is completed under my guidance.

DR. RITU JANGRA

Signature of Mentor

Date: 7/11/2022

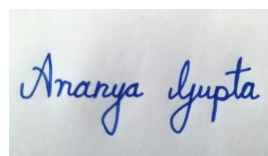
Mentor Name: Dr. Ritu Jangra

Designation: Research Associate

Organization Name: IGDTUW

UNDERTAKING REGARDING ANTI-PLAGIARISM

I, **Ananya Gupta**, hereby, declare that the material/ content presented in the report are free from plagiarism and is properly cited and written in my own words. In case, plagiarism is detected at any stage, I shall be solely responsible for it.



(Ananya Gupta)

Signature of the Student

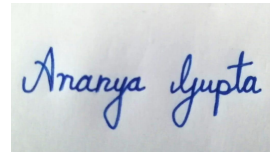
Ananya Gupta

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ACKNOWLEDGEMENT

I would like to acknowledge my mentor Dr. Ritu Jangra for her very helpful comments, support and encouragement.

Finally, I am grateful to the Department of Artificial Intelligence & Data Sciences, IGDTUW, AI Club and Centre of Excellence- Artificial Intelligence, IGDTUW for providing a healthy, supportive and understanding environment. They allowed me the freedom to explore innovative models to simplify a complex business problem. This made my project work possible without any hindrance.



(Ananya Gupta)

Signature of the Student

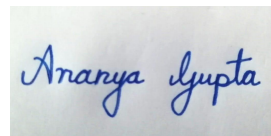
Ananya Gupta

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DECLARATION

I, **Ananya Gupta**, solemnly declare that the industrial project report, “**Major Project on Predicting rate of Employee Attrition** ” is based on my own work carried out during the course of our study under the supervision of **Dr. Ritu Jangra**. I assert the statements made and conclusions drawn are an outcome of my research work. I further certify that:

- I. The work contained in the report is original and has been done by me under the supervision of my supervisor.
- II. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this university or any other University of India or abroad.
- III. We have followed the guidelines provided by the university in writing the report.
- IV. Whenever we have used materials (text, data, theoretical analysis/equations, codes/program, figures, tables, pictures, text etc.) from other sources, we have given due credit to them in the report and have also given their details in the references.



(Ananya Gupta)

Signature of the Student

Ananya Gupta

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📎 Thu, Jun 2, 9:04 PM ⭐ ↩ ⋮

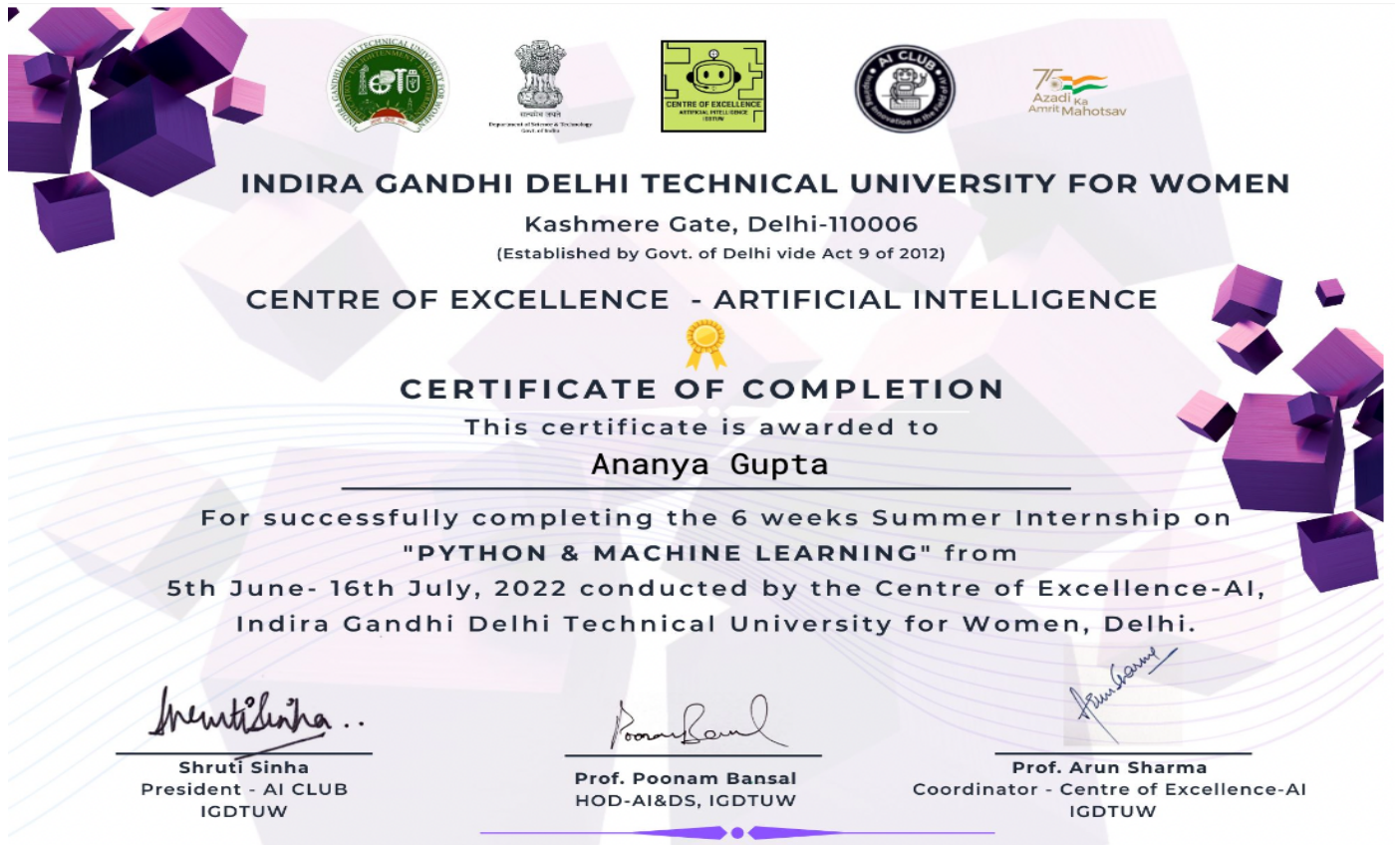
Please find the attached schedule for further details:

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AI Club IGDTUW

Internship Certificate



Manager's Feedback

Not Applicable: The mentor directly provided the certificate of completion and no softcopy of the feedback was provided.

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Job Profile

Job Detail:

1. Job position: Intern
2. Company: Department of AI & DS, AI Club and Centre of Excellence, IGDTUW

Job Responsibilities:

1. The interns were required to attend online sessions and create projects using whatever we had learnt.
2. We also had to attend sessions organised with industry professionals who told us about their work domain and experience.
3. Minor and Major Projects were to be submitted towards the end of technical sessions.
4. Additionally, Nvidia lectures on various Deep Learning and Machine Learning themes were made available to interns.
5. For successful completion of the Internship, the interns were required to produce a good research paper based on their final major projects.

About Company

About University:

Indira Gandhi Delhi Technical University for Women (IGDTUW) is a State University of the Government of Delhi. The University facilitates and promotes studies, research, technology, innovation, incubation, and extension work in emerging areas of professional education among women. The objective of the University is to foster industry-relevant research and innovations and empower the women of our country through value-based higher education making them employable, self-reliant, and responsible citizens of the country.

About Departments:

Centre of Excellence (CoE) in Artificial Intelligence (AI) at IGDTUW, established by the support of Department of Science and Technology (DST), GOI caters to the requirements of Undergraduate, Post-graduate and Doctorate programs in the domains of AI, Machine Learning and Deep Learning and various applications including Robotics, Drones, NLP and others. The centre serves as the perfect platform with the necessary hardware and software infrastructure to serve as a playground to the creative minds that solve real data-driven problems at hand.

The Department of Artificial Intelligence and Data Science (AI&DS) in the university aims to focus on a four-tiered strategy of Education, Research, Development and Innovation (ERDI Strategy) in the field of AI, Machine Learning, Deep Learning and Data Sciences. Currently, the department is running a 4 years UG program B.Tech – CSE(AI), which is specially designed to enable students to build intelligent machines, software, or applications with a cutting-edge combination of machine learning, analytics and visualization technologies.

About AI Club:

The AI Club is a technical society with the goal of spreading awareness and knowledge on topics like machine learning, deep learning and robotics. We organize hackathons, international conferences, training, mentorships etc. in collaboration with reputed multinational companies like Deloitte, Incubate IND, Amazon, Urban Company, Coding Blocks among many others.

Timeline

Below I have mentioned the Timeline for the different events which were scheduled regarding the Internship Program.

S.No.	Event	Date
1.	Last date to apply for Internship	4th June 2022
2.	Internship Dates	6th June 2022 - 18th July 2022
3.	Duration of the sessions	6th June 2022 - 18th July 2022
4.	Duration provided for the project completion	5th July 2022 - 17th July 2022
5.	Demo day of the project	18th July 2022

Mini-Tasks

During some sessions, the interns had to do hands-on practice along with the teacher for the particular session.

Minor project:

Based on the basis of Python and Data Science and Analytics, we were given a list of topics to choose from. On the chosen topic, the interns had to perform data analysis and state conclusions.

The topic chosen by me for the minor project was “Honey Production Analysis”. I downloaded the freely available dataset from Kaggle and performed data analysis on it and answered the questions based on it. At last, I also stated conclusions and suggestions for increasing the honey production demand based on my analysis of the dataset.

Link of the minor project:

<https://colab.research.google.com/drive/19jfV0XDcdHNIxUKT9LfOZYFUITAKplHr?usp=sharing>

Introduction

The internship gave students a brief overview of the core ideas in machine learning, including its mathematical foundations, programming tools, Python packages, and well-known machine learning and deep learning methods. Many real-world, industrial case walk-through applications were used to demonstrate a proper grasp of machine learning principles.

In a few of the first sessions, the interns were introduced to the Python programming language as well as the fundamentals of data science and data cleansing. We were introduced to Python libraries such as Numpy, Pandas, Matplotlib, and Seaborn. Following these sessions, the interns were required to present a small project demonstrating their ability to successfully apply the knowledge they had learned to a real-world dataset.

Additional lectures covered topics such as how models function, fundamental data exploration, missing values, categorical variables, cross-validation, underfitting and overfitting, data encoding, regression analysis, random forests, support vector machines, etc. Practical application of each of these ideas was made clear through hands-on exercises. The final tasks were assigned based on the knowledge obtained about various ML Models, their benefits, drawbacks, and features.

A research paper based on the final project was written and submitted. Various informative sessions with industry professionals were also conducted towards the end of the program.

For our final project, we made a group of three members and worked on analysing the reasons behind employee attrition using machine learning models. We did data cleaning and data preprocessing, finally preparing our dataset for training and testing. We split our dataset into 70% training and 30% testing ratio and implemented 3 different ML models namely, Logistic Regression, Support Vector Machine and Random Forest Classifier.

System Requirement Analysis

Software Requirements

Operating System: Any one of the following OS- Linux or Windows or macOS.

Programming Language: Python

Editor or IDE: Google Colab or Jupyter Notebook

Technology: Machine learning and Deep Learning

Hardware Requirements

Central Processing Unit (CPU): Intel Core i5 6th Generation processor or higher. An AMD equivalent processor will also be optimal.

Hard Disk: 245GB or above is recommended.

RAM- 8 GB minimum, 16 GB or higher is recommended.

Graphics Processing Unit (GPU) - NVIDIA GeForce GTX 960 or higher MAC's inbuilt Graphic card.

System Analysis

Feasibility Study

Employee attrition is the term used to describe the loss of employees as a result of life events like retirement, self-initiated resignations, the removal of positions, or other such occurrences.

Attrition differs from turnover in that the employer does not fill a vacancy that has occurred in a position after it has been filled. Therefore, it is crucial to identify the causes of staff attrition. What are the main criteria businesses aren't meeting, leading to low job satisfaction? Job satisfaction is a part of life satisfaction. Life happiness includes a sense of fulfilment at work. A key component of job satisfaction, which affects general life satisfaction, is the kind of work one does.

Economic Feasibility

It is of utmost importance to know the reasons behind the loss of employees from a company as they are the prominent people who know the working areas, tactics and knowledge about the growth of a company. The companies face loss of manpower, experience and revenue. So, a project based on preventing employee attrition by finding out the causes of attrition will contribute to the company's overall growth and profit. The project did not require any large amount of expense or costly resources. The project is made with resources freely available on Youtube, Kaggle and other articles and blogs. The project mostly comprised working on the real-world dataset.

Implementation Feasibility

We used 70% of our dataset for training and 30% for testing. The accuracy of the best model is displayed in the form of a percentage which is the final result of our project. We got an accuracy of 86.39%. Hence the implementation of our project is feasible.

Work Description

A. Dataset

For our project, we have taken a sample employee attrition dataset from Kaggle. The dataset contains 1470 records and 35 fields including categorical and numeric features. Each record in the employee data set represents a single employee's information and each field in the record represents a feature of that particular employee.

B. Pre-processing of dataset

Data pre-processing is one of the most critical aspects of machine learning, and is estimated to cover about 60% of the time and energy spent on a data science project. Therefore, it should be carried out with the most sincerity and attention, as it forms the backbone of the further stages of the project process.

1) Data Selection:

First, Data Selection is done:

The data relevant to the project is selected from the initial dataset. For example, characteristics like 'EmployeeCount', 'EmployeeNumber', 'Over18', 'StandardHours' etc are removed, which are insignificant and redundant. Then, Binary Encoding, Ordinal Encoding and One-hot encoding is carried out to transform the qualitative variables into quantitative variables and the categorical data into numbers so that the machine learning model could work. The original dataset in fact contained several variables with textual values ("BusinessTravel", "Department", "EducationField"). Therefore we applied transcoding to transform the n values of a class into numeric variables, from 0 to n minus 1. For example, the "BusinessTravel" variable, consisting of 3 values, was transformed as follows:

- (i) The "Non-Travel" value was mapped with the value 0,
- (ii) The "Travel-Frequently" value was mapped with the value 1,
- (iii) The "Travel-Rarely" value was mapped with the value 2.

2) Data Exploration:

In order to observe the characteristics of all the variables, we considered the following variables: count, unique, top, frequency, mean, standard deviation (std), minimum and maximum values (min/max), 25%/50%/75% percentile. .

We plotted the correlation matrix heat map. It graphically illustrates correlations among all variables: the bluish fields represent no correlation, while the relative intensity of the dark

blue and white colours represents an increase in correlation. In particular, dark blue reveals a positive or direct correlation (the variation of one characteristic directly affects the other) and white reveals a negative or indirect correlation (the variation of a characteristic inversely affects that of the other).

3) Test dataset and training dataset:

Separating data into test datasets and training datasets is an important part of evaluating data mining models. The test data set contains all the required data for data prediction and training data set contains all irrelevant data. Here we have 441 records in test dataset and 1029 records in training dataset. We apply data classification and data prediction on the test dataset of 441 records.

C. Model Building

The modelling process consists of selecting the best models, based on various machine learning techniques. In this case, various predictive models were used such as those based on random forest, logistic regression and SVM. The goal is to identify the best classifier for the analysed problem. Each classifier must therefore be trained on the featured set and the classifier with the best classification results is used for prediction. The classification algorithms taken into consideration are:

- (i) Logistic Regression classifier,
- (ii) Support Vector Machine classifier,
- (iii) Random Forest classifier.

Random Forest Classifier: The random forest is a classification algorithm consisting of many decision trees. It uses bagging and feature randomness when building each individual tree to try to create an uncorrelated forest of trees whose prediction by committee is more accurate than that of any individual tree.

Logistic Regression Classifier: The Logistic Regression is a classification technique used in machine learning. It uses a logistic function to model the dependent variable. The dependent variable is dichotomous in nature, i.e. there could only be two possible classes (for example: either the cancer is malignant or not).

Support Vector Machine Classifier: SVM is a supervised machine learning algorithm which can be used for classification or regression problems. It performs classification by finding the hyperplane that completely separates the vector into two non-overlapping classes. The vectors that define the hyperplane are the support vectors.

Work Outcome

➤ Minor Project Outcome:

- Got to learn about python libraries like NumPy, pandas, and matplotlib.
- I stated conclusions and suggestions for increasing the honey production demand based on my analysis of the dataset.
- I was able to conclude that the total honey production has declined over the years and the value of production per pound has increased.
- Reason:
There was a continuous decrease in the yield per colony over the years.

➤ Major Project Outcome:

- Learnt about preprocessing of data
- Implemented three ML models on the given dataset.
- Calculated the accuracy, f1 score of each model.
- The models Random Forest and Logistic Regression gave f1 scores of 0.241 and 0.473. These models gave an accuracy of 86.39% and 84.35%.
- From the analysis of the dataset and calculating accuracies and f1 score, we found SVM has the highest accuracy, meaning it guesses correctly **86.39%** of the predictions.
- We wrote a research paper at the end which is still under review by our mentor.
- Certificate was awarded after successful completion of the Internship.

Conclusion

According to F1 score and Accuracy, the best model out of Logistic Regression, Random Forest and SVM is SVM.

Reasons to choose the SVM Model:

- (i) It has the highest F1 score among SVM, Logistic Regression and Random Forest.
- (ii) It has the highest accuracy among SVM, Logistic Regression and Random Forest.

Employee attrition is not a completely unavoidable problem for organizations. However, they can address it at the right time. Firstly, they need to compute the attrition rate so they can fully understand the issue and the causes behind it. They can then come up with a well-planned, data-driven process to reach out to new job candidates, engage them with different strategies, and retain them for a longer period of time.

Link to the Major Project:

<https://colab.research.google.com/drive/15jM5z2DHNMkTbrD-qXYkXb9h0YOrlJJ?usp=sharing>

Plagiarism

Not Provided till now:

No plagiarism report for the Internship Report was provided by any of the mentors or the company till now. So, I could not attach one.

Research Paper

Title: Predicting rate of employee attrition using machine learning algorithms

Authors: Ananya Gupta, Pari Khaitan and Sakshi Sihag

Conference: N.A.

Status: Under review by the Faculty.

Description:

Employees are the most valuable assets of the organization. Their presence adds value to the organization in terms of both quality and quantity. When employees leave the organization, they carry with them invaluable tacit knowledge which is often the source of competitive advantage for the business. Thus, business leaders sometimes have to face huge setbacks due to the loss of talented employees. Irrespective of the kind of industry or the structure of the business, attrition is a common problem in every organization which not only hampers production but also results in heavy long-run costs and loss of goodwill to the organization.

In this paper, an attempt has been made to analyse the publicly available employee attrition dataset using suitable models, study them and propose the best model on the basis of their prediction accuracies of employee attrition.

Link to the research paper:  [Research_Paper_on_Employee_Attrition.pdf](#)

References

Blogs and Articles:

- <https://www.indeed.com/hire/c/info/employee-attrition>
- <https://www.hibob.com/blog/measuring-attrition/#:~:text=How%20can%20you%20calculate%20your,100%20to%20get%20the%20percentage.>
- <https://towardsdatascience.com/understanding-confusion-matrix-a9ad42dcfd62>
- <https://www.analyticsvidhya.com/blog/2021/11/employee-attrition-prediction-a-comprehensive-guide/>
- https://www.cksvim.edu.in/wp-content/uploads/2020/07/2059_cksvim-nikita_sip.pdf
- <https://corporatefinanceinstitute.com/resources/management/attrition-2/>